PROPER PUSHFORWARD OF A COHERENT SHEAF OF DELTA-DIMENSION AT MOST K COMMUTES WITH THE K-DIMENSIONAL CYCLE CLASS

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file spaces-chow.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: a support-and-scheme-theoretic-image reduction to closed immersions followed by a component comparison and a shrink to the finite affine case

Editorial note. The label lemma-cycle-push-sheaf names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- Cohomology of Spaces, proper pushforward of a coherent module is coherent, and every coherent sheaf on a good algebraic space is the pushforward of a coherent module on a closed subspace that is its support
- The closed immersion case of the cycle class computation for a pushforward, and the identification of the k-dimensional part of a cycle class with a sum over the irreducible components of delta-dimension k
- Spaces over Fields, finiteness in codimension 1 near a generic point of a component, and Chow Homology's cycle pushforward lemma for schemes, used as the final reduction step

Source reference. Original lemma 02R6: sources/stacks-chow/tags.txt, source lines 3613–3613. Prerequisite pointer invoked at proof line(s) 1117; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 02QY: sources/stacks-spaces-chow/pinned-original-source.tex, source lines 762–762. Prerequisite pointer invoked at proof line(s) 1043; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0EE3: sources/stacks-spaces-chow/pinned-original-source.tex, source lines 521–521. Prerequisite pointer invoked at proof line(s) 1080; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0EE6: sources/stacks-spaces-chow/pinned-original-source.tex, source lines 626–626. Prerequisite pointer invoked at proof

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line(s) 1099; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 02QL: sources/stacks-spaces-chow/pinned-original-source.tex, source lines 264–264. Prerequisite pointer invoked at proof line(s) 1022; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 07UG: sources/stacks-chow/tags.txt, source lines 9971–9971. Prerequisite pointer invoked at proof line(s) 1047; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 08AR: sources/stacks-chow/tags.txt, source lines 10541–10541. Prerequisite pointer invoked at proof line(s) 1039; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original definition 082Y: sources/stacks-chow/tags.txt, source lines 10268–10268. Prerequisite pointer invoked at proof line(s) 1053; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 04WP: sources/stacks-chow/tags.txt, source lines 6367–6367. Prerequisite pointer invoked at proof line(s) 1067; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 04XZ: sources/stacks-chow/tags.txt, source lines 6412–6412. Prerequisite pointer invoked at proof line(s) 1068; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 082Z: sources/stacks-chow/tags.txt, source lines 10269–10269. Prerequisite pointer invoked at proof line(s) 1064; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 04NX: sources/stacks-chow/tags.txt, source lines 6095–6095. Prerequisite pointer invoked at proof line(s) 1069; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 0AD2: sources/stacks-chow/tags.txt, source lines 13075–13075. Prerequisite pointer invoked at proof line(s) 1106; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original proposition 06NH: sources/stacks-chow/tags.txt, source lines 8536–8536. Prerequisite pointer invoked at proof line(s) 1112; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *In Situation 02QL let $f : X \to Y$ be a proper morphism of good algebraic spaces over B.*

(1) *Let $Z \subset X$ be a closed subspace with $\dim_\delta(Z) \leq k$. Then*

$$f_*[Z]_k = [f_*\mathcal{O}_Z]_k.$$

(2) *Let $\mathcal{F}$ be a coherent sheaf on X such that $\dim_\delta(Supp(\mathcal{F})) \leq k$. Then*

$$f_*[\mathcal{F}]_k = [f_*\mathcal{F}]_k.$$

Note that the statement makes sense since $f_*\mathcal{F}$ and $f_*\mathcal{O}_Z$ are coherent $\mathcal{O}_Y$-modules by Cohomology of Spaces, Lemma 08AR.

Proof. Part (1) follows from (2) and Lemma 02QY. Let $\mathcal{F}$ be a coherent sheaf on X. Assume that $\dim_\delta(\mathrm{Supp}(\mathcal{F})) \leq k$. By Cohomology of Spaces, Lemma 07UG there exists a closed immersion $i : Z \to X$ and a coherent $\mathcal{O}_Z$-module $\mathcal{G}$ such that $i_*\mathcal{G} \cong \mathcal{F}$ and such that the support of $\mathcal{F}$ is Z. Let $Z' \subset Y$ be the scheme theoretic image of $f|_Z : Z \to Y$, see Morphisms of Spaces, Definition 082Y. Consider the commutative diagram

$$\begin{array}{ccc} Z & \xrightarrow{i} & X \\ {\scriptstyle f|_Z}\downarrow & & \downarrow{\scriptstyle f} \\ Z' & \xrightarrow{i'} & Y \end{array}$$

of algebraic spaces over B. Observe that $f|_Z$ is surjective (follows from Morphisms of Spaces, Lemma 082Z and the fact that $|f|$ is closed) and proper (follows from Morphisms of Spaces, Lemmas 04WP, 04XZ, and 04NX). We have $f_*\mathcal{F} = f_*i_*\mathcal{G} = i'_*(f|_Z)_*\mathcal{G}$ by going around the diagram in two ways. Suppose we know the result holds for closed immersions and for $f|_Z$. Then we see that

$$f_*[\mathcal{F}]_k = f_*i_*[\mathcal{G}]_k = (i')_*(f|_Z)_*[\mathcal{G}]_k = (i')_*[(f|_Z)_*\mathcal{G}]_k = [(i')_*(f|_Z)_*\mathcal{G}]_k = [f_*\mathcal{F}]_k$$

as desired. The case of a closed immersion follows from Lemma 0EE3 and the definitions. Thus we have reduced to the case where $\dim_\delta(X) \leq k$ and $f : X \to Y$ is proper and surjective.

Assume $\dim_\delta(X) \leq k$ and $f : X \to Y$ is proper and surjective. For every irreducible component $Z \subset Y$ with generic point η there exists a point $\xi \in X$ such that $f(\xi) = \eta$. Hence $\delta(\eta) \leq \delta(\xi) \leq k$. Thus we see that in the expressions

$$f_*[\mathcal{F}]_k = \sum n_Z[Z], \quad \text{and} \quad [f_*\mathcal{F}]_k = \sum m_Z[Z].$$

whenever $n_Z \neq 0$, or $m_Z \neq 0$ the integral closed subspace Z is actually an irreducible component of Y of δ-dimension k (see Lemma 0EE6). Pick such an integral closed subspace $Z \subset Y$ and denote η its generic point. Note that for any $\xi \in X$ with $f(\xi) = \eta$ we have $\delta(\xi) \geq k$ and hence ξ is a generic point of an irreducible component of X of δ-dimension k as well (see Lemma 0EE6). By Spaces over Fields, Lemma 0AD2 there exists an open subspace $\eta \in V \subset Y$ such that $f^{-1}(V) \to V$ is finite. Since η is a generic point of an irreducible component of $|Y|$ we may assume V is an affine scheme, see Properties of Spaces, Proposition 06NH. Replacing Y by V and X by $f^{-1}(V)$ we reduce to the case where Y is affine, and f is finite. In particular X and Y are schemes and we reduce to the corresponding result for schemes, see Chow Homology, Lemma 02R6 (applied with $S = Y$). □